ICE-4107: Microwave Engineering

Bangladesh University of Professionals

Department of Information & Communication Technology

ICE-4107: Microwave Engineering

Course Objectives:

1. Use transmission line analysis techniques.
2. Analyze waveguides structures propagating TE, TM or TEM modes, e.g., rectangular or circular waveguides, coaxial lines, surface wave lines, strip line, and microstrip lines.
3. Understand the concepts of microwave network analysis.
4. Design various impedance matching devices such as stub tuners and multi-stage quarter wave transformers.
5. Analyze and design microwave resonators.
6. Design microwave power dividers and directional couplers.

ICE-4107: Microwave Engineering: Course Contents

Frequency spectrum: VHF, UHF and microwave frequency ranges, microwave, advantages and applications.

Overview of Maxwell's and Helmholtz Equations, Plane wave and its solution, Poynting's theorem, Plane wave reflection for normal and oblique incidence.

Transmission Line: Voltage and current in ideal transmission lines, reflection, transmission, standing wave, impedance transformation, Smith Chart: impedance matching and lossy transmission lines.

Waveguides: General formulation, modes of propagation and losses in parallel plate, rectangular and circular waveguides. Strip-line, Micro strips.

Resonators: structures and characteristics. Rectangular Resonant Cavities, Energy storage, losses, Q.

Microwave circuit and devices: microwave tubes, transmit time and velocity modulation, Klystron, Multi-cavity Klystron, Oscillator, Magnetron.

Radiation: Small current element, radiation resistance, Radiation pattern and properties, Hertzian and halfwave Dipoles, Beam solid angle, radiation intensity, directivity, effective aperture.

Network Analysis: Scattering Matrices and Multiport Analysis Techniques.

ICE-4107: Microwave Engineering

Course Outcomes:

1. Gain knowledge and understanding of microwave analysis methods.
2. Be able to apply analysis methods to determine circuit properties of passive/active microwave devices.
3. Know how to model and determine the performance characteristics of a microwave circuit or system using computer aided design methods.
4. Have knowledge of basic communication link design; signal power budget, noise evaluation and link carrier to noise ratio.
5. Have knowledge of how transmission and waveguide structures and how they are used as elements in impedance matching and filter circuits.

References:

1. Microwave Devices and Circuits, (third edition) by Samuel Y. Liao.
2. Foundations for Microwave Engineering – Robert E. Collin.;
3. Microwave Engineering - M. Pozar.
4. Field Theory of Guided Waves by Robert E. Collin
5. Antennas and Wave Propagation by John D Kraus, R. J Marhefka and A S Khan
6. Engineering Electromagnetics (Sixth Edition) by William H. Hayt

Lecture 01

Conduct of the Course

- Class Lectures: 28 (Max.)
- Lecture Materials: Recommended books and slides
- Attendance is MUST

→ 4 Class tests 10 (Best 3 will be counted)

→ Assignments 10

→ Attendance 10

→ Midterm 20

→ Final exam 50

Lecture 01

Grading

Numerical Grade	Letter Grade		Grade Point
80% and above	A+	(A Plus)	4.00
75% to < 80%	A	(A Regular)	3.75
70% to < 75%	A-	(A Minus)	3.50
65% to < 70%	B+	(B Plus)	3.25
60% to < 65%	B	(B Regular)	3.00
55% to < 60%	B-	(B Minus)	2.75
50% to < 55%	C+	(C Plus)	2.50
45% to < 50%	C	(C Regular)	2.25
40% to < 45%	D	-	2.00
< 40%	F	-	0.00

Class Attendance

Attendance	Marks
95% and above	10.0
90% to 94%	9.0
85% to 89%	8.0
75% to 84%	7.0
Less than 75%	Dis-collegiate

Lecture 01

Lecture Plan

Lecture	Topics	Rmks / Discussion points
Lec 1	Frequency spectrum, VHF, UHF and microwave frequency ranges, microwave, advantages and applications	1. Microwaves: frequency band, 2. EM waves, 3. General applications of microwaves
Lec 2	Overview of Maxwell's and Helmholtz Equations, Plane wave and its solution,	1. Maxwell's equation 2. Helmholtz equations 3. Plane wave and its solution
Lec 3	Classtest 01, Poynting's theorem, Plane wave reflection for normal and oblique incidence.	1. Poynting's theorem 2. Plane wave reflection of different incidence
Lec 4	Transmission Line: Voltage and current in ideal transmission lines, reflection, transmission,	1. Transmission line parameters, 2. General line equation, 3. lossless line, 4. $\lambda/4$ line,
Lec 5	standing wave, impedance transformation, Smith Chart: impedance matching and lossy transmission lines.	1. Standing waves , VSWR, 2. reflection coefficient, 3.stub matching (single and double), 4. skin effect

Lecture 01

Lecture Plan

Lec 7	Modes of propagation and losses in parallel plate, rectangular and circular waveguides.	1. Rectangular waveguide: structure , cut off wavelength , 2. group and phase velocities, characteristic wave impedance , 3. TE ,TM modes, field patterns , examples, S Parameters basics
Lec 8	Strip-line, Micro strips, Resonators: structures and characteristics.	1. Basics of strip line 2. Micro strip line 3. Principles of resonators
Lec 9	Class Test 03, Resonant Cavities: Energy storage, losses, Q. Microwave circuit and devices: microwave tubes, transmit time and velocity modulation,	1. Resonant Cavities: Energy storage, losses, Q. 2. Microwave tubes, transmit time and velocity modulation,
Lec 10	Klystron, Multi-cavity Klystron, Oscillator, Magnetron	1. Microwave tubes amplifiers: Klystron - Two cavity and multi cavity, 2. Travelling Wave Tube 3. Microwave tubes oscillators: Reflex klystron, Magnetron,

Lecture 01

Lecture Plan

Lec 11	Radiation: Small current element, radiation resistance, Radiation pattern and properties, Hertzian and halfwave Dipoles,	1. Small current element, 2. radiation resistance, 3. Radiation pattern and properties, 4. Hertzian and halfwave Dipoles,
Lec 12	Class Test 04, Beam solid angle, radiation intensity, directivity, effective aperture.	1. Beam solid angle, 2. Antenna parameters: radiation intensity, directivity, effective aperture.
Lec 13	Antenna: Introduction to antenna arrays and their design, radiation from a dipole antenna, antenna temperature,	1. Array antenna and their design 2. Dipole antenna 3. Antenna parameter: effect of temperature
Lec 14	Mono-pole, horn, rhombic and parabolic reflectors.	1. Mono-pole, 2. horn, rhombic and 3. parabolic reflectors.

Lecture 01

Lecture Plan

Lec 15	Class Test 05, Microwave Communication systems: Types, Friis power transmission formula,	1. Microwave Communication systems, 2. Types, 3. Friis power transmission formula,
Lec 16	Microwave transmitters and receivers. RADAR systems: Application,	1. Microwave transmitters and receivers. 2. Introduction to RADAR systems: Application,
Lec 17	Class Test 06, Radar equation, range, Types of Radar, Polarization, noise, interference,	1. Radar equation, range, 2. Types of Radar: MTI, 3. Polarization, noise, interference,
Lec 18	Atmospheric and ground effects. Other applications of microwave, microwave oven etc	1. Atmospheric and ground effects. 2. Other applications of microwave, 3. microwave oven etc
Lec 19	Presentation /Viva Voce	
Lec 20	Presentation /Viva Voce	
Lec 21	Semester Final Examination	

Microwave Frequencies

- Microwave is a form of electromagnetic radiation with wavelengths ranging from 30 cm to 1 mm (1 to 300 GHz).
- However, microwave really indicates the wavelengths in the micron ranges. Microwaves are "small", compared to waves used in typical radio broadcasting
- This means microwave frequencies are up to infrared and visible-light regions. In this revision, microwave frequencies refer to those from 1 GHz up to 10^6 GHz.

Properties of Microwave Frequencies

- Microwaves are the waves that radiate electromagnetic energy with shorter wavelength.
- Microwaves are not reflected by Ionosphere.
- Microwaves travel in a straight line and are reflected by the conducting surfaces.
- Microwaves are easily attenuated within shorter distances.
- Microwave currents can flow through a thin layer of a cable.

Microwave Frequencies

- In August 1969 the United States Department of Defense, Office of Joint Chiefs of Staff, by message to all services, directed the use of a new frequency band breakdown as shown in Table 0-1.

TABLE 0-1 U.S. MILITARY MICROWAVE BANDS

Designation	Frequency range in gigahertz
P band	0.225– 0.390
L band	0.390– 1.550
S band	1.550– 3.900
C band	3.900– 6.200
X band	6.200– 10.900
K band	10.900– 36.000
Q band	36.000– 46.000
V band	46.000– 56.000
W band	56.000–100.000

Microwave Frequencies

- On May 24, 1970, the Department of Defense, USA adopted another band designation for microwave frequencies as listed in Table 0-2.

TABLE 0-2 U.S. NEW MILITARY MICROWAVE BANDS

Designation	Frequency range in gigahertz	Designation	Frequency range in gigahertz
A band	0.100–0.250	H band	6.000– 8.000
B band	0.250–0.500	I band	8.000– 10.000
C band	0.500–1.000	J band	10.000– 20.000
D band	1.000–2.000	K band	20.000– 40.000
E band	2.000–3.000	L band	40.000– 60.000
F band	3.000–4.000	M band	60.000–100.000
G band	4.000–6.000		

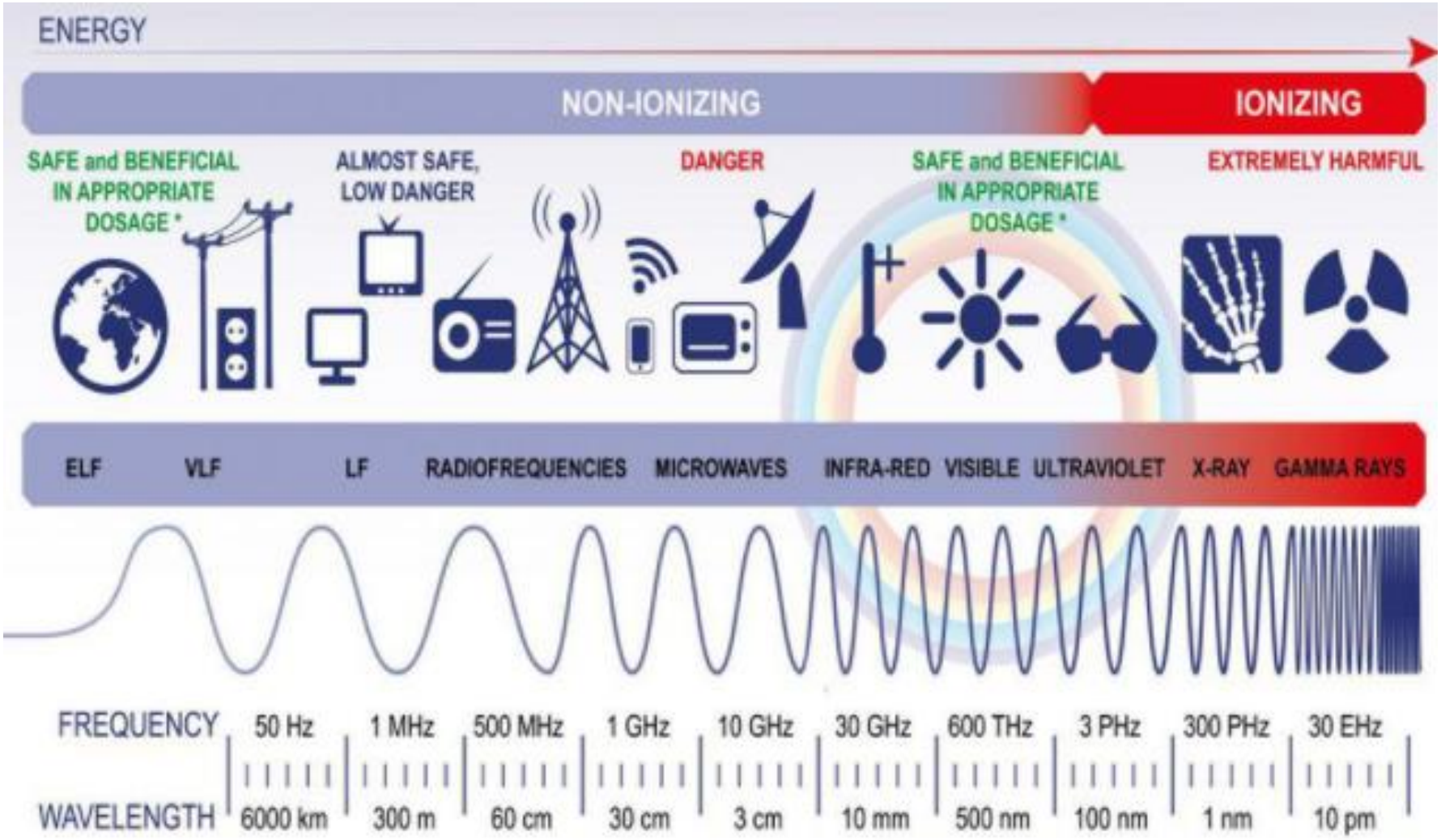
Microwave Frequencies

- The Institute of Electrical and Electronics Engineers (IEEE) recommended new microwave band designations as shown in Table 0-3 for comparison.

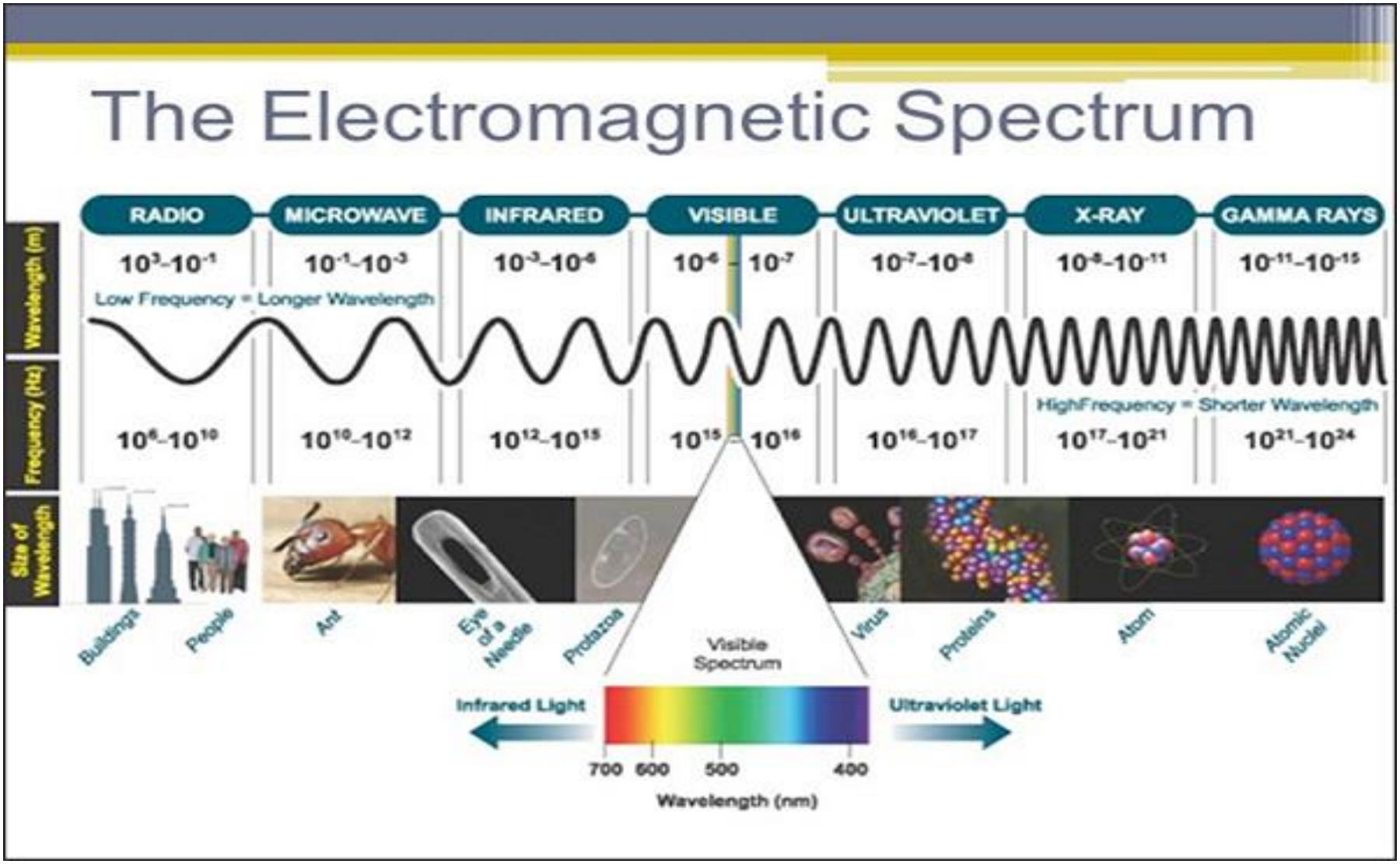
TABLE 0-3 IEEE MICROWAVE FREQUENCY BANDS

Designation	Frequency range in gigahertz
HF	0.003– 0.030
VHF	0.030– 0.300
UHF	0.300– 1.000
L band	1.000– 2.000
S band	2.000– 4.000
C band	4.000– 8.000
X band	8.000– 12.000
Ku band	12.000– 18.000
K band	18.000– 27.000
Ka band	27.000– 40.000
Millimeter	40.000–300.000
Submillimeter	>300.000

Microwave Frequencies



- Microwave Frequencies



Microwave Frequencies: Advantages

- Supports larger bandwidth and hence more information is transmitted.

  For this reason, microwaves are used for point-to-point communications.

- More antenna gain is possible.
- Higher data rates are transmitted as the bandwidth is more.
- Antenna size gets reduced, as the frequencies are higher.
- Low power consumption as the signals are of higher frequencies.
- Effect of fading gets reduced by using line of sight propagation.
- Provides effective reflection area in the radar systems.

Microwave Frequencies: Advantages

- Satellite and terrestrial communications with high capacities are possible.
- Low-cost miniature microwave components can be developed.
- Effective spectrum usage with wide variety of applications in all available frequency ranges of operation.

Microwave Frequencies: Disadvantages

- Cost of equipment or installation cost is high.
- They are hefty and occupy more space.
- Electromagnetic interference may occur.
- Variations in dielectric properties with temperatures may occur.
- Inherent inefficiency of electric power.

Microwave Frequencies: Applications

There are many Industrial, Scientific, Medical and Domestic Applications of Microwaves. The great example of Application of Microwaves is 'Microwave Oven' which we uses in our daily life. Following are the other main application areas of Microwaves:

Wireless Communications

- For long distance telephone calls
- Bluetooth
- WIMAX operations
- Outdoor broadcasting transmissions
- Broadcast auxiliary services
- Remote pickup unit
- Studio/transmitter link

Microwave Frequencies: Applications

Military and Radar

- Radars to detect the range and speed of the target.
- SONAR applications
- Air traffic control
- Weather forecasting
- Navigation of ships
- Minesweeping applications
- Speed limit enforcement

Microwave Frequencies: Applications

Industrial Uses

- Vulcanizing rubber
- Analytical chemistry applications
- Drying and reaction processes
- Processing ceramics
- Polymer matrix
- Surface modification
- Chemical vapor processing
- Powder processing
- Sterilizing pharmaceuticals
- Chemical synthesis

- **Microwave Frequencies: Applications**

Medical Applications

- Monitoring heartbeat
- Lung water detection
- Tumor detection
- Regional hyperthermia
- Therapeutic applications
- Local heating
- Angioplasty
- Microwave tomography
- Microwave Acoustic imaging

Thank You